CSC 311 – Design and Analysis of Algorithms

Project I – Convex Hull

Brute Force, Divide-and-Conquer (QuickHull), and Graham Scan

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1. Introduction

The convex hull of a set of points P in the Euclidean plane is the smallest convex polygon that contains all points in the set. This project implements three algorithms to compute the convex hull: Brute Force, QuickHull, and Graham Scan. Each algorithm reads input points, computes the hull, displays intermediate steps, and plots the result.

2. Data Representation

Input points are read using Reader.getData(). Points are stored as strings and converted into java.awt.Point objects. Hull points are stored in ArrayLists for drawing.

3. Algorithm Descriptions

3.1 Brute Force (Gift Wrapping version)

Starts at the lowest point, repeatedly selects next point with maximum angle. Complexity: O(nh), worst O(n^2).

3.2 QuickHull (Divide & Conquer)

Splits points by leftmost/rightmost, recursively finds farthest points. Average O(n log n), worst O(n^2).

3.3 Graham Scan

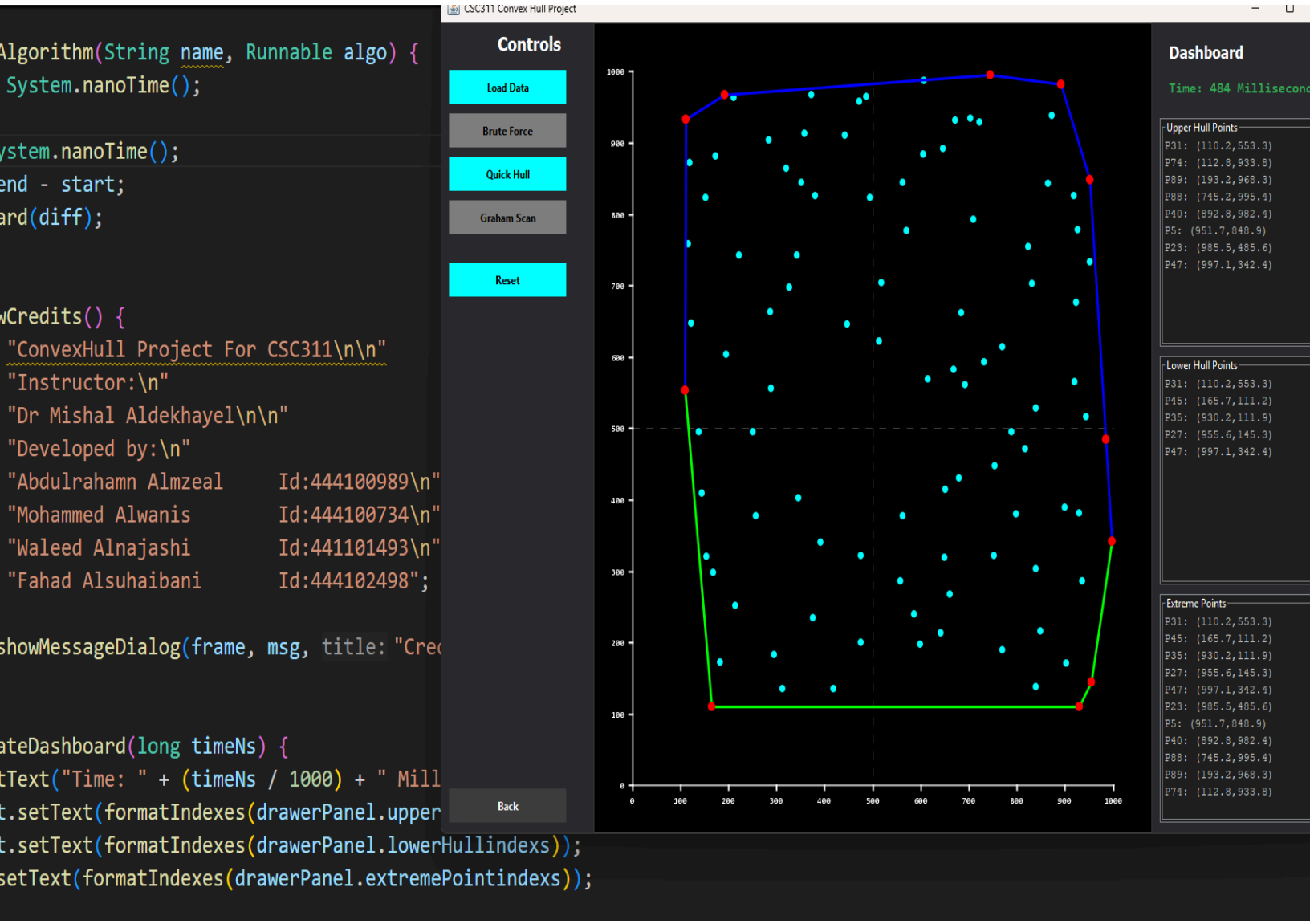
Sorts by angle, uses stack to maintain left turns. Complexity O(n log n).

4. Time Complexity Comparison

Brute Force: O(n^2) QuickHull: O(n log n) average Graham Scan: O(n log n)

5. Sample Run (datapoints1.txt)

This is simple run in bruteForce Points displayed, hull printed in the GUI, Blue and Green hull drawn in GUI.



6. Conclusion

Graham Scan was fastest and most stable. QuickHull performs well on average. Brute force is simplest but slowest. GUI visualization assisted understanding of algorithm behavior.